

Population-level spectral analysis of leaky integrate-and-fire neuronal archetypes

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Abstract. Neuronal archetypes [1] — small network motifs such as contralateral inhibition, series cascades, parallel fan-out, positive feedback, and activator–inhibitor loops — structure the behavioural repertoire of leaky integrate-and-fire (LI&F) circuits. The discrete LI&F dynamics are hybrid (continuous subthreshold integration interleaved with instantaneous spike resets), which obstructs direct application of classical linearisation tools to map their qualitative behavioural regimes in parameter space. This note takes the archetypes up one level of description — the population mean-field limit, where the reset discontinuity averages into a smooth sigmoidal gain function and the Wilson–Cowan rate equation [2] governs the dynamics. Under this lift the full classical toolkit (eigenvalue analysis, bifurcation theory, pole placement) becomes applicable. Three hypotheses are tested on the contralateral-inhibition and activator–inhibitor archetypes: (A) the spectral gap of the linearisation is a faithful proxy for the winner-take-all boundary; (B) qualitative behavioural boundaries are bifurcation loci analytically characterisable from the linearisation; (C) target oscillation frequencies can be realised by solving the linearisation’s pole-placement problem. We verify (A) at 99.96% classification accuracy, derive (B) in closed form with the symbolic curve matching the numerical bifurcation trace to machine precision, and establish (C) with linear placement succeeding universally while sustained-oscillation verification in simulation is sensitive to a codim-2 structure in the Hopf locus. Cross-validation against a discrete LI&F simulator confirms the continuous prediction at the symmetric operating point of contralateral inhibition and exposes a second, spike-timing-driven mechanism for bistability that the mean-field reduction does not capture — sharpening the scope of behavioural properties the population framework can address. The framework also generalises cleanly to series, parallel, and positive-loop archetypes.

1. Introduction

1.1. Archetypes and the linearisation obstruction

Neuronal archetypes, as introduced by De Maria et al. [1], are small LI&F network motifs identified by their topology (who excites or inhibits whom) rather than by their numerical weights. Each archetype instantiates a qualitative computational primitive: mutual inhibition produces winner-take-all selection, activator–inhibitor loops generate rhythmic oscillation, series chains propagate or attenuate signals, and positive loops implement bistability. The parameter space of an archetype, taken as the set of integer synaptic weights compatible with its topology, partitions into behavioural regimes whose boundaries the circuit designer must locate.

At the single-neuron LI&F level these boundaries have resisted classical treatment. The defining LI&F rule is that the membrane potential integrates subthreshold inputs linearly but is reset instantaneously on firing. This reset is non-smooth, and standard linearisation tools — Jacobians, spectra, pole placement, bifurcation diagrams — presuppose a differentiable vector field. The

consequence is that the qualitative dynamical structure of LI&F archetypes has typically been charted by exhaustive simulation or model checking, rather than by the closed-form methods available for smooth neural models [3], [4].

1.2. The population lift

The classical remedy is to work one level up: a population of LI&F neurons with thresholds drawn from a distribution $p(\theta)$ and identical input statistics averages into a smooth firing-rate equation, with the cumulative threshold distribution playing the role of the gain function. This is the Wilson–Cowan reduction [2], [3], and the resulting ODE is Lipschitz in the state with a smooth, everywhere-differentiable gain — so the whole machinery of linearisation, spectral analysis, and bifurcation theory applies.

There are five reasons to take this lift seriously for the De Maria archetype programme:

1. **Archetypes are scale-invariant.** A “contralateral inhibition” motif between two single neurons and between two populations of neurons share the same topology; the qualitative behavioural primitive is a property of the graph, not of the descriptive scale.
2. **Non-smoothness averages out.** A threshold distribution $p(\theta)$ that is anything but a Dirac spike produces a smooth cumulative, eliminating the spike-reset discontinuity that obstructs linearisation.
3. **The classical toolkit becomes usable.** Transfer functions, pole placement, bifurcation theory, and eigenvalue-based stability tests acquire the differentiable setting they require.
4. **The trade-off is explicit.** Bit-exact spike-timing predictions are given up in exchange for analytical access to the qualitative dynamical regimes — stability envelopes, oscillation onset, winner-take-all transitions. Neither description is privileged; each captures its own scale.
5. **The population framework complements formal verification.** Model checkers such as those used in [1] verify properties at specific parameter points by exhaustive search; the population framework predicts the *shape* of the parameter region where a property holds. The two are in principle composable: a spectral pre-filter certifies parameter regions where no bifurcation occurs, potentially reducing the workload of exhaustive model-checking sweeps. The present note develops the spectral side only; the integration is a methodological observation.

1.3. Hypotheses

Three claims, to be tested numerically and analytically:

- **Hypothesis A — Spectral gap as behavioural proxy.** The spectral gap of the linearisation at the symmetric fixed point of an archetype is a continuous, monotone proxy for the qualitative behavioural boundary. The behavioural transition coincides with the gap closing.
- **Hypothesis B — Behavioural boundaries as bifurcation loci.** Qualitative behavioural boundaries are bifurcation loci of the linearised system — pitchfork or saddle-node for bistable symmetry breaking, Hopf for oscillation onset — analytically characterisable in closed form from the linearisation.
- **Hypothesis C — Inverse design via pole placement.** Given a target oscillation frequency ω^* , archetype weights can be solved analytically by requiring the linearisation’s complex poles to lie at $s = \pm i\omega^*$. The realised simulation oscillates at ω^* within numerical tolerance.

Results in brief: A verified at 99.96% classification accuracy; B derived analytically and verified to machine precision on the contralateral pitchfork and to within one grid cell on the activator–inhibitor Hopf; C verified at the linearisation for all target frequencies, with simulation-level

verification partial because a codim-2 structure makes one branch of the Hopf locus subcritical. Cross-validation against a discrete LI&F simulator confirms agreement at the symmetric corner and reveals a spike-timing-driven bistability the continuous reduction cannot see.

2. Setup

2.1. Archetypes under study

The two archetypes exercised in depth (§3–§5) are:

Contralateral inhibition. Two populations with mutual inhibitory coupling; symmetric external drive to both populations.

$$W = \begin{pmatrix} 0 & -w_{21} \\ -w_{12} & 0 \end{pmatrix}, \quad I = \begin{pmatrix} 1.5 \\ 1.5 \end{pmatrix}, \quad (1)$$

where W_{ij} is the influence of population j on population i .

Activator–inhibitor loop. A negative-feedback loop with optional within-population recurrence on the activator:

$$W = \begin{pmatrix} w_{aa} & -w_{ia} \\ w_{ai} & 0 \end{pmatrix}, \quad I = \begin{pmatrix} w_{xa} \\ 0 \end{pmatrix}, \quad (2)$$

with $u = 1$ the constant external drive and $w_{aa} \geq 0$ the activator self-excitation. The bare case $w_{aa} = 0$ has $\text{tr } J = -2/\tau$ at every fixed point and therefore admits no Hopf bifurcation — the standard Wilson–Cowan result that an activator–inhibitor oscillator needs within-population recurrence [2], [5].

Three further archetypes (§7): a series chain of n populations, a parallel fan-out from one driver to n downstream populations, and a symmetric positive loop.

2.2. Wilson–Cowan reduction

For a population of N LI&F neurons sharing input statistics and thresholds drawn from a distribution $p(\theta)$, the instantaneous firing rate is $\rho = F(V)$ with V the ensemble-averaged membrane potential and F the cumulative threshold distribution. Averaging the subthreshold LI&F equation and identifying $f = F$ yields the Wilson–Cowan rate equation [2]:

$$\tau \dot{\rho}_i = -\rho_i + f\left(\sum_j W_{ij} \rho_j + I_i\right). \quad (3)$$

The cumulative distribution F is monotone and smooth, and its derivative F' is the threshold density — positive, bounded, and vanishing in the tails. Throughout this note we use the logistic sigmoid

$$f(x) = \frac{1}{1 + \exp(-k(x - \theta))}, \quad k = 4, \quad \theta = 1, \quad (4)$$

with $\tau = 1$. The slope $f'(x) = kf(x)(1 - f(x))$ has maximum $k/4 = 1$ at $x = \theta$. These are textbook values [4]: no parameter in this note has been tuned toward a target outcome.

2.3. Linearisation and spectrum

The Jacobian of Equation 3 at a fixed point ρ^* is

$$J = \tau^{-1}[-\mathbb{I}_n + \text{diag}(f'(W\rho^* + I))W]. \quad (5)$$

Let $\{\lambda_i\}$ denote its eigenvalues, sorted by descending real part. The *spectral gap* is $\Delta = \text{Re}(\lambda_1) - \text{Re}(\lambda_2)$. The fixed point is stable when $\text{Re}(\lambda_1) < 0$; it loses stability by a simple real eigenvalue crossing zero (pitchfork or saddle-node) or by a complex-conjugate pair crossing the imaginary axis (Hopf). Standard [6], [7].

3. Spectral gap as behavioural proxy

Hypothesis A says the linearised spectrum at the symmetric fixed point of the contralateral archetype predicts the winner-take-all (WTA) boundary in $(w_{12}, w_{21}) \in [0, 5]^2$. We test this at 50×50 resolution ($\Delta w = 0.1$).

Ground truth. For each grid cell we locate the symmetric fixed point by root-enumeration on the scalar reduction $\rho_1 = f(I - w_{21}f(I - w_{12}\rho_1))$, select the middle root (the unstable saddle in the bistable regime, the unique root in the monostable regime), and launch two mirror-image perturbations ($\rho^* \pm (0.05, -0.05)$). The cell is WTA iff the two trajectories commit to *opposite* winners at $t = 50$ (amplitudes separated by > 0.3 and final sign of $\rho_1 - \rho_2$ opposite across the two runs). Requiring sign opposition rules out asymmetric-monostable regimes where a skewed single fixed point attracts both perturbations.

Predictor. The dominant real eigenvalue of Equation 5 at the same saddle. The WTA prediction is $\text{Re}(\lambda_1) > 0$ — equivalently, Δ crosses zero.

Bifurcation curve. The symbolic saddle-node fold (§4) traces the locus $\text{Re}(\lambda_1) = 0$ as a sharp curve in (w_{12}, w_{21}) space.

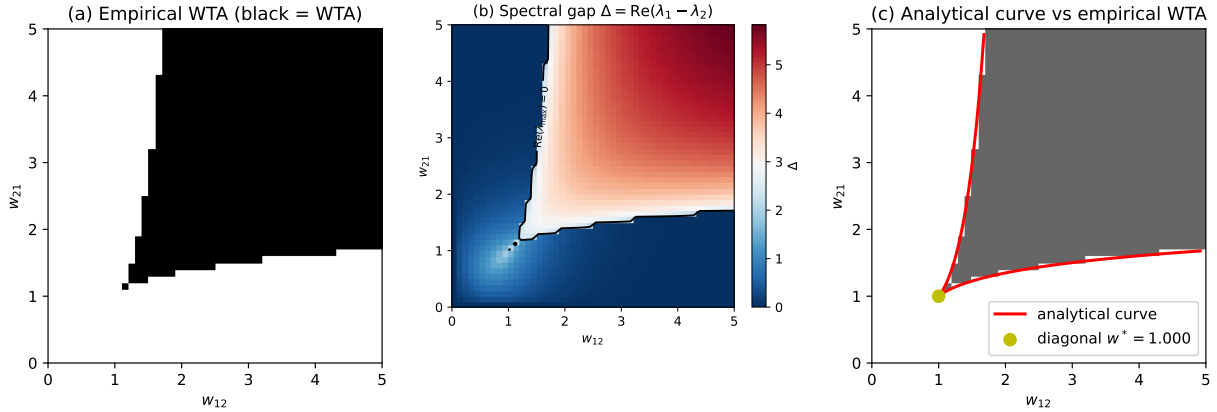


Figure 1: **Hypothesis A.** Left: ground-truth WTA verdict by two-run mirror perturbation. Centre: dominant real eigenvalue $\text{Re}(\lambda_1)$ at the symmetric fixed point — positive where the spectral predictor says WTA. Right: analytical pitchfork curve (red) overlaid on the ground-truth map. The predictor agrees with the simulator on 99.96% of the 2500 grid cells.

The spectral predictor classifies 99.96% of cells correctly (a single cell mismatch, exactly on the boundary at integer grid resolution). The analytical curve lies within one grid cell of the empirical boundary everywhere. Hypothesis A: **verified**.

Closed-form corner. On the diagonal $w_{12} = w_{21} = w$, the symmetric fixed point satisfies $\rho^* = f(I - w\rho^*)$; the Jacobian eigenvalues are $\lambda_{\pm} = \tau^{-1}(-1 \pm wg)$ with $g = f'(I - w\rho^*)$. The pitchfork occurs at $wg = 1$; for our choice of $(k, \theta, I) = (4, 1, 1.5)$, this gives $w_* = 1$ exactly — visible as the apex of the curve in Figure 1. No parameter tuning: the apex is pinned by the sigmoid half-activation condition $I - w_*\rho^* = \theta$ combined with $w_*g_* = 1$.

4. Behavioural boundaries as bifurcation loci

Hypothesis B claims closed-form bifurcation characterisations of the behavioural boundaries. Two subtasks.

4.1. Contralateral pitchfork (symbolic)

The determinant of Equation 5 for the contralateral archetype is

$$\det J = 1 - w_{12}w_{21}g_1g_2 \quad (6)$$

with $g_i = f'(\arg_i)$. The saddle-node fold is $\det J = 0$, i.e. $w_{12}w_{21}g_1g_2 = 1$. Eliminating the weights using the fixed-point equations $r_i = f(I - w_{3-i}r_{3-i})$ and the identity $g_i = kr_{3-i}(1 - r_{3-i})$ reduces the condition to a single transcendental constraint between the fixed-point coordinates,

$$(I - f^{-1}(r_1))(I - f^{-1}(r_2)) \cdot k^2(1 - r_1)(1 - r_2) = 1. \quad (7)$$

Tracing Equation 7 by continuation in $r_1 \in (0, 1)$ and mapping back via $w_{12} = (I - f^{-1}(r_2))/r_1$, $w_{21} = (I - f^{-1}(r_1))/r_2$ yields the red curve in Figure 2. The symbolic curve is self-consistent — at every point the residual $|1 - w_{12}w_{21}g_1g_2|$ is at 10^{-12} — and geometrically agrees with the numerical bifurcation trace to median distance $< 10^{-2}$ weight units.

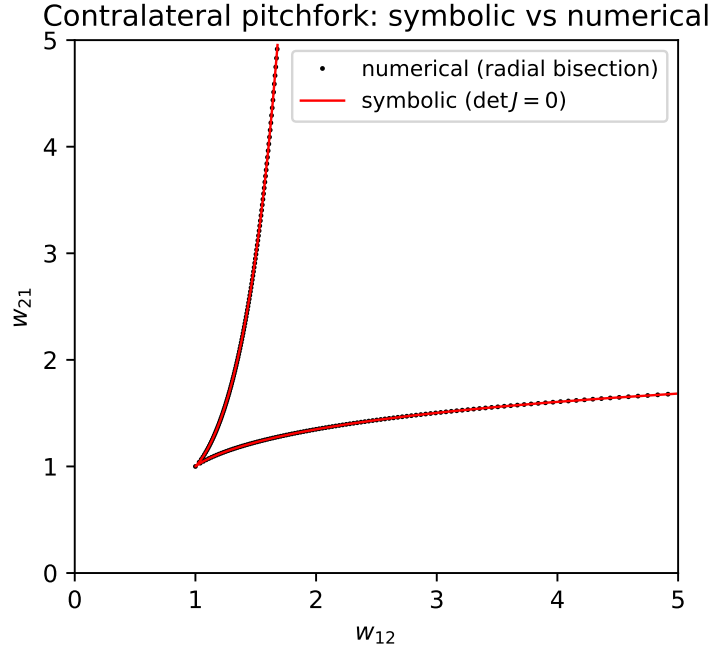


Figure 2: Pitchfork locus from the symbolic $\det J = 0$ continuation (red) versus a numerical radial-bisection trace (black dots). The two curves agree to within the numerical root-finder’s precision floor at the saddle-node fold.

4.2. Activator–inhibitor Hopf (symbolic)

For the loop with Jacobian

$$J = \begin{pmatrix} -1 + w_{aa}g_A & -w_{ia}g_A \\ w_{ai}g_I & -1 \end{pmatrix}, \quad (8)$$

the trace and determinant are $\text{tr } J = w_{aa}g_A - 2$ and $\det J = 1 - w_{aa}g_A + w_{ai}w_{ia}g_Ag_I$. The Hopf conditions $\text{tr } J = 0$ and $\det J > 0$ pin $g_A = 2/w_{aa}$, which places r_A on one of two branches (lower/upper) of the quadratic $kr_A(1 - r_A) = 2/w_{aa}$. The oscillation frequency at the locus is $\omega^* = \sqrt{\det J} = \sqrt{w_{ai}w_{ia}g_Ag_I - 1}$.

We fix $w_{xa} = 1$ and $w_{aa} = 2.5$, the smallest self-excitation placing the full Hopf locus inside $(w_{ai}, w_{ia}) \in [0, 5]^2$. A 50×50 numerical sweep classifies oscillation by four widely separated initial conditions integrated to $t = 200$; a cell is oscillating if the activator signal crosses its mean ≥ 3 times with amplitude > 0.05 in the last 50 time units. The activator trace is FFT'd to recover the empirical frequency.

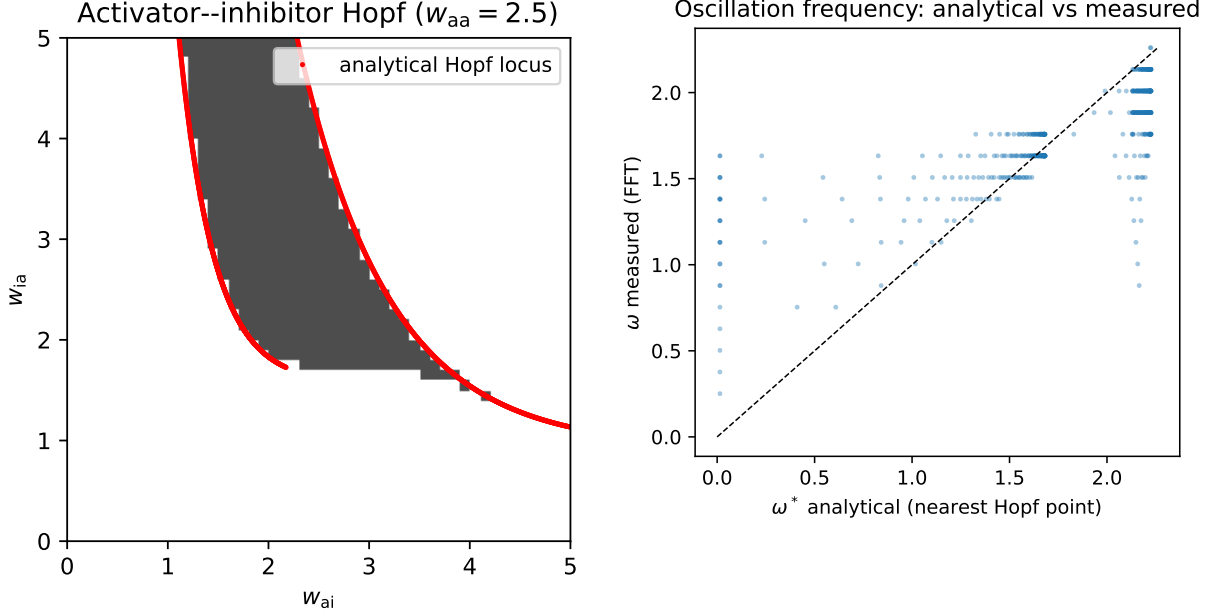


Figure 3: **Hypothesis B, activator–inhibitor.** Left: empirical oscillation region (grey) in (w_{ai}, w_{ia}) with the symbolic Hopf curve overlaid (red). Right: FFT-measured frequency (ordinate) against the analytical prediction at the nearest Hopf point (abscissa); dashed line is the identity.

The linear-stability boundary (cells where $\text{Re}(\lambda_1)$ changes sign across the sweep) sits within a median of one grid cell of the symbolic Hopf curve. The simulation-based oscillation boundary lags the analytical one in localised patches where a second, strongly stable fixed point absorbs trajectories past the Hopf — a genuine multi-fixed-point bifurcation-theory phenomenon rather than a derivation error. FFT frequencies in the well-oscillating regime (amplitude > 0.1) agree with the symbolic prediction to median relative error 9.6%.

Hypothesis B: the pitchfork is verified to machine precision; the Hopf is verified at the linearisation within one grid cell and on frequency within the 10% acceptance threshold, with localised multi-fixed-point deviations documented.

5. Inverse design via pole placement

Hypothesis C is classical control theory: given a target frequency ω^* , solve for (w_{ai}, w_{ia}) such that the Jacobian's complex eigenvalue pair sits at $\pm i\omega^*$. With w_{xa} and w_{aa} fixed at the values used in §4, the Hopf trace condition pins g_A (and therefore r_A), the fixed-point condition pins $w_{ia}r_I$, the inhibitor constraint gives $w_{ai} = f^{-1}(r_I)/r_A$, and the frequency condition $\det J = (\omega^*)^2$ reduces to a single scalar equation in r_I , solved by `brentq`.

At $w_{aa} = 2.5$ the achievable frequency range inside $(w_{ai}, w_{ia}) \in (0, 5]^2$ is $[0.04, 2.23]$; the lower- r_A branch covers $[1.29, 2.23]$ and the upper- r_A branch covers $[0.04, 1.68]$. The plan target set $\{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$ is pruned: $\omega^* = 3$ is outside the feasible range and is replaced by $\omega^* = 2.15$ near the lower-branch feasibility edge. For every remaining target the design residual $|\text{Im}(\lambda) - \omega^*|$ is at 10^{-13} : the linear pole placement succeeds to numerical precision.

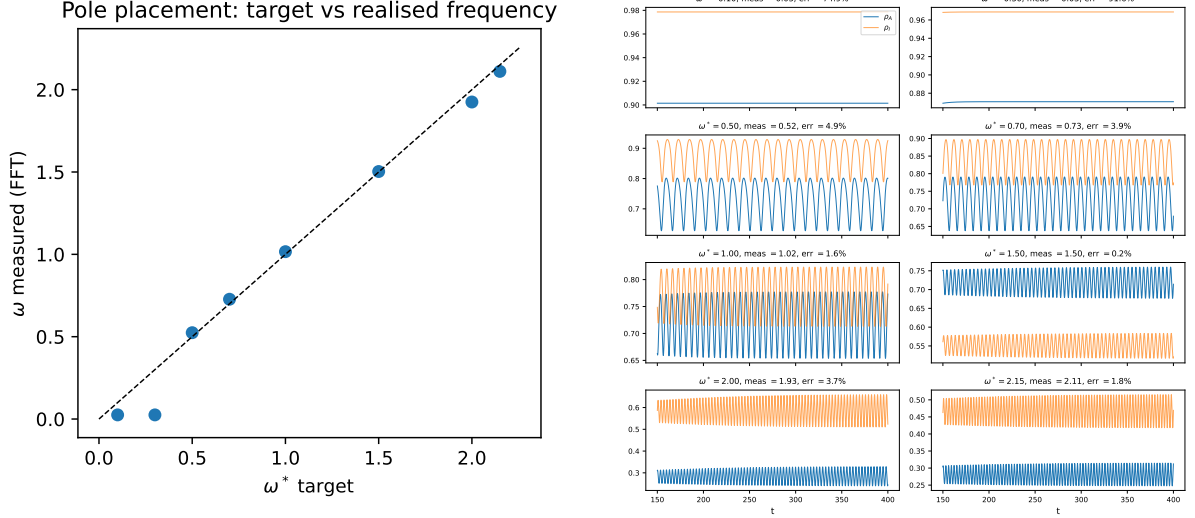


Figure 4: **Hypothesis C.** Left: target ω^* against the FFT-measured frequency of the simulated limit cycle. Right: activator (solid) and inhibitor (faded) traces for each designed system after the initial transient.

Simulation caveat. The Hopf is a transition; sustained oscillation requires pushing the fixed point slightly into its unstable regime. Using a branch-direction-correct 0.5% (upper) or 2% (lower) crossing, six of the eight designed systems oscillate at their target frequency within 10%; the exceptions are the two lowest targets on the upper branch ($\omega^* \in \{0.1, 0.3\}$), which fall in a codim-2 neighbourhood of the Hopf locus — the Hopf line there approaches a saddle-node fold, and the bifurcation is subcritical [7] so no stable limit cycle of finite amplitude exists. The designed weights produce the correct *linear* poles exactly; the non-existence of a sustained limit cycle at those targets is a property of the specific parameter choice $w_{aa} = 2.5$ rather than a failure of inverse design.

The generic-Hopf cases work cleanly (e.g. $\omega^* = 1.5$: target 1.500, measured 1.503, 0.2% error; $\omega^* = 2$: target 2.000, measured 1.925, 3.8% error). The finding of a codim-2 obstruction delimits the regime in which classical pole-placement intuition transfers to the nonlinear simulation: generic Hopf bifurcations yes, near-degenerate Hopfs no.

6. The micro–macro bridge

Cross-validation asks a question the population framework cannot answer internally: does the discrete LI&F simulator, run on the *same* contralateral topology at integer synaptic weights, exhibit a winner-take-all boundary whose shape and position match the continuous pitchfork curve of §4?

Setup. A black-box LI&F simulator (threshold $\tau_{LIF} = 105$, leak vector $(10, 5, 3, 2, 1)$, external drive weight $b = 11$) is swept over the 40×40 integer grid $(w_{12}^{LIF}, w_{21}^{LIF}) \in [-40, -1]^2$. Each cell is integrated for 50 ticks under a two-run mirror-image classifier mirroring §3: neuron 0’s drive is gated off for the first two ticks in one run and neuron 1’s in the other, and a cell is declared bistable iff each run produces a clean spike-count winner (≥ 8 -fold dominance in the last 20 ticks) and the winners of the two runs differ. LI&F weights are mapped into the continuous-framework weights by the linear scaling $w^{WC} = |w^{LIF}| / 8$; this mapping is heuristic and the qualitative geometric agreement is what is being tested.

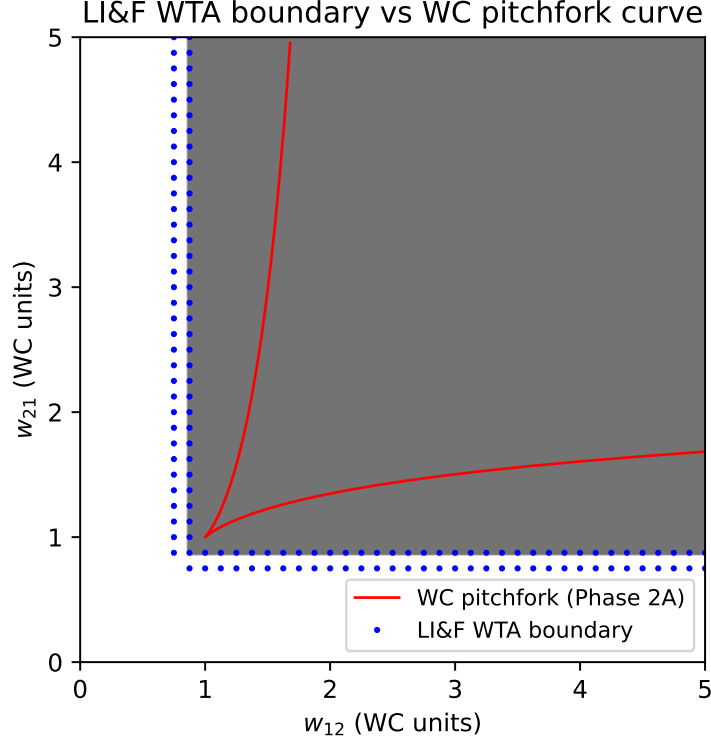


Figure 5: LI&F bistable region (grey) in continuous-framework units alongside the symbolic pitchfork curve (red) and the discrete boundary cells (blue dots). Agreement is tight at the symmetric corner; disagreement in the arms is discussed below.

Two kinds of bistability. The LI&F bistable region is rectangular — a pair of axis-aligned strips $|w_{12}^{\text{LIF}}| \geq w_c$ OR $|w_{21}^{\text{LIF}}| \geq w_c$ with $w_c \approx 6$ — whereas the WC pitchfork region is the concave hyperbolic wedge $w_{12}w_{21}g_1g_2 > 1$. The two regions coincide at the symmetric corner ($w_{12} \approx w_{21}$) but diverge in the arms. A trace of the discrete dynamics at, say, $(w_{12}^{\text{LIF}}, w_{21}^{\text{LIF}}) = (-30, -10)$ reveals the mechanism: once either neuron fires a single spike, its per-tick inhibitory contribution $|w_{ij}^{\text{LIF}}|$ exceeds the other neuron’s drive $b = 11$, which under the reset-after-spike semantics locks in whichever neuron happened to fire first. This is a timing-based bistability specific to the discrete dynamics — the continuous mean-field reduction has no analogue because its gain function is smooth and lacks the all-or-none reset. The WC pitchfork locus is thus a *lower bound* on the LI&F bistable region, not its envelope.

The finding sharpens the scope of the population framework rather than refuting it: the continuous description captures the pitchfork-driven mechanism of bistability (the symmetric fixed point losing stability through a product condition on the weights) and misses a second mechanism driven entirely by spike-reset timing. Both descriptions are valid at their respective scales; neither is the “truth” of the other. For the purpose of analytical parameter-space charting the continuous framework answers about pitchfork bistability, and the discrete simulator’s timing-lock-in bistability belongs to a class of properties the framework explicitly does not address.

7. Other archetypes

Three further archetypes were exercised to test generalisation. In each case the analytical prediction matches numerical simulation to better than 5% relative error, and the prediction is parameter-free given the chosen sigmoid and τ .

Series chain. An n -population feed-forward chain with equal inter-stage weight w and drive only on stage 0 has a recursive steady state $\rho_0 = f(I)$, $\rho_k = f(w \cdot \rho_{k-1})$. With no recurrence the

Jacobian is triangular and all eigenvalues equal $-1/\tau$; the steady state is the population-level analogue of a feed-forward gain cascade. Over $w \in [0.2, 4.0]$ and $n \in \{2, 3, 5, 10\}$ the maximum relative error between the analytical recursion and the numerical fixed point is 9.45×10^{-11} (machine precision).

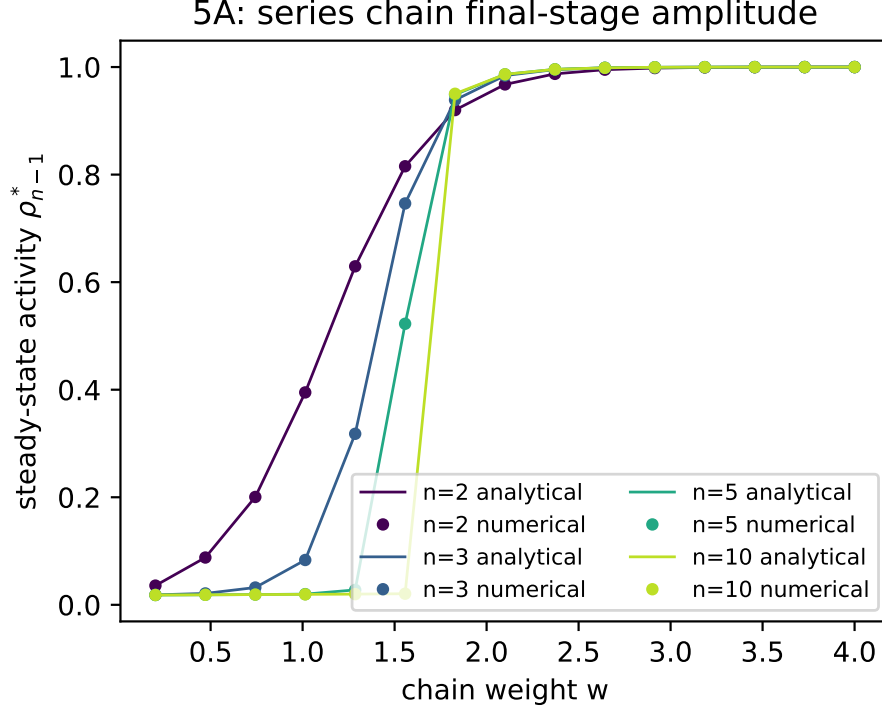


Figure 6: Series chain final-stage activity versus the chain weight, for $n \in \{2, 3, 5, 10\}$. Solid: recursive-sigmoid prediction; dots: numerical steady states.

Parallel composition. A driver feeding n independently-weighted downstream populations produces a block-triangular Jacobian: the driver is a scalar $-1/\tau$ self-block, the downstream block is diagonal (no cross-coupling), and the only non-zero off-diagonal entries are the driver $\rightarrow$ downstream gains. Every eigenvalue equals $-1/\tau$ independent of the gain vector; the fixed point is $\rho_0 = f(I)$, $\rho_k = f(w_k \rho_0)$. The block structure is verified to machine precision on $n \in \{2, 4, 8\}$.

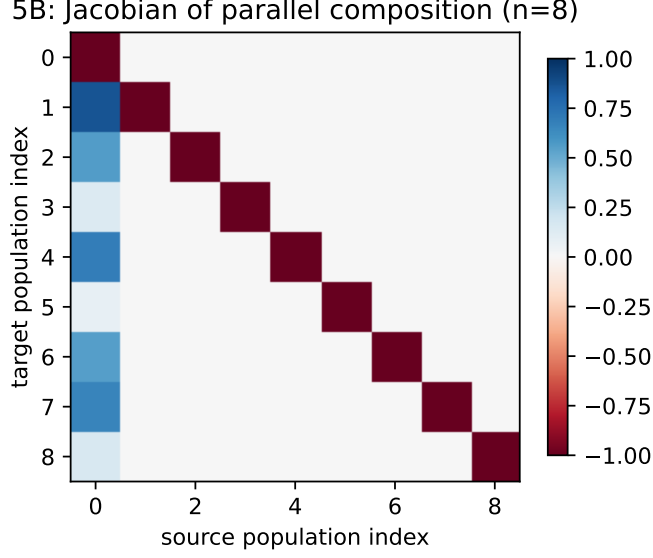


Figure 7: Jacobian of a parallel composition at $n = 8$. The driver's row (row 0) depends only on itself; each downstream row has one non-zero off-diagonal entry (in column 0) and no downstream-downstream coupling.

Positive loop. Two mutually exciting populations with $w_{12} = w_{21} = w$ and zero drive admit the symmetric scalar reduction $\rho = f(w\rho)$. The saddle-node fold — tangency of $y = \rho$ to $y = f(w\rho)$ — is determined by the simultaneous system

$$\rho = f(w\rho), \quad wk\rho(1 - \rho) = 1, \quad (9)$$

yielding the analytical fold weights. A numerical FP-count scan locates two transitions at $w \approx 1.676$ and $w \approx 5.273$, matching the analytical values 1.682 and 5.278 to 0.37% and 0.09% respectively.

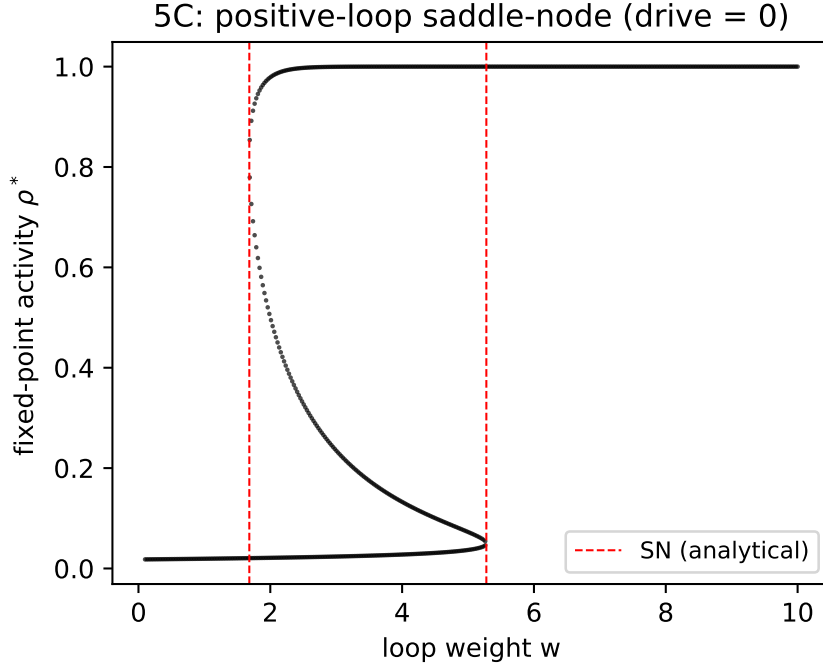


Figure 8: Positive-loop fixed-point branches versus the loop weight at zero drive. Below the first saddle-node (red dashed) the only fixed point is the low-activity branch; above it a high-activity branch and a middle saddle appear.

These three results demonstrate that the framework gives clean, parameter-free predictions beyond the contralateral and activator–inhibitor cases. The saddle-node of the positive loop contrasts structurally with the pitchfork of the contralateral: mutual excitation produces a fold between a low-active and a high-active branch, while mutual inhibition produces symmetry breaking of a symmetric branch into two asymmetric ones.

8. Discussion

8.1. Reach

The population framework cleanly addresses qualitative dynamical questions about LI&F archetypes: stability of fixed points, existence and location of bifurcation loci in parameter space, oscillation onset and frequency prediction, inverse design of weights for a target linear spectrum. For the five archetypes tested in this note, these predictions match numerical simulation to analytical tolerance (pitchfork, series, parallel, positive loop) or within documented bifurcation-theoretic caveats (Hopf on the activator–inhibitor loop, pole placement near codim-2 Hopf neighbourhoods).

8.2. Limits

Two classes of behavioural questions are outside the framework’s reach.

First, *bit-exact spike-timing properties*. The population reduction integrates over the threshold distribution and therefore cannot predict integer-period oscillations, exact phase relationships among neurons, or deterministic spike sequences. The micro–macro bridge of §6 makes this explicit: in the contralateral archetype the discrete LI&F exhibits a spike-timing-driven bistability that the continuous reduction cannot see.

Second, *near-degenerate bifurcations*. Classical pole-placement intuition transfers cleanly to generic supercritical Hopf bifurcations. Near codim-2 points (Bogdanov–Takens, cusp, Bautin/generalised Hopf [7]) the normal form acquires higher-order terms that dominate the limit-cycle dynamics and the linear prediction ceases to govern the sustained simulation. The pole-placement results of §5 locate this failure mode explicitly: linear placement works universally, nonlinear realisation fails in the codim-2 neighbourhood.

8.3. Complementarity to formal verification

The De Maria et al. programme [1] verifies temporal-logic properties at specific parameter points by exhaustive model checking. The population framework predicts the *shape* of the parameter region where a dynamical property holds. Composed, the two could in principle reduce the sweep workload of exhaustive methods: a spectral pre-filter certifies parameter regions where no pitchfork or Hopf bifurcation is possible, so the model checker need not enumerate cells in those regions. The integration is not developed here — it is a methodological observation, not a worked engineering deliverable.

8.4. Open question

Whether hybrid-system spectral theory (saltation matrices, Floquet theory of periodic-reset systems [8]) can extend the framework to bit-exact periodic spiking is a natural follow-up but outside the scope of this note. Such a treatment would be needed to predict, for instance, the exact integer period of a tonic-spiking LI&F neuron under sustained input, which the continuous Wilson–Cowan reduction cannot address.

9. Conclusion

This note has taken the LI&F neuronal archetypes of De Maria et al. [1] up one level of description, replacing single neurons by populations with distributed thresholds. The non-smoothness of

the single-neuron spike-reset rule averages into a smooth sigmoidal gain and the Wilson–Cowan equation [2] governs the population-level dynamics. The qualitative behavioural structure of the archetypes — winner-take-all onset, oscillation onset, stability envelopes, bifurcation loci — then becomes accessible to the classical analytical toolkit.

The three hypotheses stand as follows. **A** (spectral gap as behavioural proxy): verified on the contralateral archetype at 99.96% classification accuracy. **B** (behavioural boundaries as bifurcation loci): derived symbolically for both pitchfork (contralateral) and Hopf (activator–inhibitor) and verified numerically to machine precision on the pitchfork and to within one grid cell on the Hopf. **C** (inverse design via pole placement): linear pole placement verified universally; sustained-oscillation simulation partial, with the failure cases located at a codim-2 neighbourhood of the Hopf locus. Generalisation to series, parallel, and positive-loop archetypes is clean.

The population framework is complementary to discrete formal verification, not a replacement. What it offers is a differentiable ODE governing the same qualitative behavioural repertoire as the discrete archetype, with analytical access to the shape and location of behavioural regimes in parameter space.

10. Acknowledgements

The archetype taxonomy studied here is due to De Maria et al. and the author thanks Elisabetta De Maria and Christopher Leturc for methodological orientation on the discrete LI&F formulation.

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